

Variables

A variable is a named memory location used to store a value.

```
int age;  
age = 25;  
int age = 25;
```

Different Types of Variables

```
int age = 25;  
double salary = 45000.50;  
char grade = 'A';  
boolean passed = true;  
String name = "Ashok";
```

Changing the Value

```
int marks = 50;  
System.out.println(marks);  
marks = 80;  
System.out.println(marks);
```

```
50  
80
```

Local Variable

A variable declared **inside a method, constructor, or block** is called a **local variable**.

```
class Demo {  
    public static void main(String[] args) {  
        int age = 25;  
        System.out.println(age);  
    }  
}  
25
```

Important:

A local variable **does not get a default value**.

```
public static void main(String[] args) {  
    int age;  
    System.out.println(age); // ERROR  
}
```

Instance Variable

A variable declared **inside a class but outside methods** is called an **instance variable**.

```
class Student {  
    String name;  
    int age;  
    public static void main(String[] args) {
```

```

        Student s = new Student();
        System.out.println(s.name);
        System.out.println(s.age);
    }
}
null
0

```

Static Variable

A variable declared using the static keyword is called a **static variable**.

```

class Student {
    static String college = "Anna University";
    public static void main(String[] args) {
        System.out.println(Student.college);
    }
}

```

final Variable

A variable declared using final cannot be changed after initialization.

```
final int MAX = 100;
```

```
System.out.println(MAX);
```

This is not allowed:

```
MAX = 200; // ERROR
```

Variable Naming Rules

Rule	Valid Example	Invalid Example
A variable name must begin with a letter, underscore (_), or dollar sign (\$).	age, _count, \$price	2age
A variable name cannot begin with a digit.	num1	1num
A variable name can contain letters, digits, underscores, and dollar signs.	student1, total_marks	student@
Spaces are not allowed in variable names.	studentAge	student Age
Special characters such as @, #, %, and - are not allowed.	total_amount	total-amount
Java keywords cannot be used as variable names.	myClass	class, int, static
Variable names are case-sensitive.	age, Age, AGE	—

A variable name must be a valid Java identifier and cannot be a reserved keyword.	value	public
A variable name can be of any length, subject to compiler and implementation limits.	studentRegistrationNumber	—
A variable must be declared with a data type before it can be used.	int marks = 90;	marks = 90; (if undeclared)

A Complete Example

```

class Student {

    // Instance variables
    String name;
    int age;

    // Static variable
    static String college = "Anna University";

    // Final variable
    final int MAX_MARKS = 100;

    public static void main(String[] args) {

        // Local variables
        int marks = 85;
        boolean passed = true;

        Student s = new Student();

        s.name = "Arun";
        s.age = 20;

        System.out.println("Name      : " + s.name);
        System.out.println("Age       : " + s.age);
        System.out.println("College   : " + Student.college);
        System.out.println("Marks    : " + marks);
        System.out.println("Passed   : " + passed);
        System.out.println("Max Marks : " + s.MAX_MARKS);
    }
}

```

Name : Arun
 Age : 20
 College : Anna University
 Marks : 85
 Passed : true
 Max Marks : 100

Summary

Variable type	Declared where?	Belongs to	Default value?
Local	Inside method/block	Method/block	☐ No
Instance	Inside class, outside method	Object	☐ Yes
Static	Inside class with static	Class	☐ Yes
Final	With final	Depends on where declared	Value cannot be changed